THE OCEANS: CURRENTS, TIDES, SALINITY / INDIAN OCEAN AND IOD

OCEAN-CURRENT FORCE CHAIN → DENSITY-OVERTURNING CHAIN → GYRE COMPARISON → UPWELLING-FOOD WEB → TIDE GEOMETRY → LOCAL TIDAL RANGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 OCEAN-CURRENT FORCE CHAIN

OCEAN-CURRENT FORCE CHAIN | WIND STRESS | SURFACE MOMENTUM | TURNING BY HEMISPHERE | PRESSURE GRADIENT | GEOSTROPHIC TENDENCY | BASIN + FRICTION | REALISED CURRENT PATH

WIND STRESS → surface momentum → CORIOLIS → turning by hemisphere → PRESSURE GRADIENT → geostrophic tendency → BASIN + FRICTION → realised current path

DEFINITION / COMPONENTS
- WIND STRESS → surface momentum
- CORIOLIS → turning by hemisphere

TRIGGER / MECHANISM
- PRESSURE GRADIENT → geostrophic tendency
- BASIN + FRICTION → realised current path

EFFECT / INTERVENTION
- WIND STRESS → surface momentum
- CORIOLIS → turning by hemisphere

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRESSURE GRADIENT → geostrophic tendency; BASIN + FRICTION → realised current path.

01

STAGE 01 DENSITY-OVERTURNING CHAIN

DENSITY-OVERTURNING CHAIN | DENSITY INCREASE | DENSITY DECREASE | DENSE WATER | SINKING AND SPREADING | MIXING + TOPOGRAPHY | OVERTURNING PATHWAYS

COOLING / EVAPORATION → density increase → FRESHWATER / WARMING → density decrease → DENSE WATER → sinking and spreading → MIXING + TOPOGRAPHY → overturning pathways

DEFINITION / COMPONENTS
- COOLING / EVAPORATION → density increase
- FRESHWATER / WARMING → density decrease

TRIGGER / MECHANISM
- DENSE WATER → sinking and spreading
- MIXING + TOPOGRAPHY → overturning pathways

EFFECT / INTERVENTION
- COOLING / EVAPORATION → density increase
- FRESHWATER / WARMING → density decrease

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DENSE WATER → sinking and spreading; MIXING + TOPOGRAPHY → overturning pathways.

02

STAGE 02 GYRE COMPARISON

GYRE COMPARISON | NORTH SUBTROPICS | CLOCKWISE TENDENCY | SOUTH SUBTROPICS | ANTICLOCKWISE TENDENCY | WESTERN LIMB | NARROW, SWIFT, WARM | EASTERN LIMB

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
NORTH SUBTROPICS → clockwise tendency	WESTERN LIMB → narrow, swift, warm	NORTH SUBTROPICS → clockwise tendency
SOUTH SUBTROPICS → anticlockwise tendency	EASTERN LIMB → broad, cool, upwelling-prone	SOUTH SUBTROPICS → anticlockwise tendency

DEFINITION / COMPONENTS
- NORTH SUBTROPICS → clockwise tendency
- SOUTH SUBTROPICS → anticlockwise tendency

TRIGGER / MECHANISM
- WESTERN LIMB → narrow, swift, warm
- EASTERN LIMB → broad, cool, upwelling-prone

EFFECT / INTERVENTION
- NORTH SUBTROPICS → clockwise tendency
- SOUTH SUBTROPICS → anticlockwise tendency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WESTERN LIMB → narrow, swift, warm; EASTERN LIMB → broad, cool, upwelling-prone.

03

STAGE 03 UPWELLING-FOOD WEB

UPWELLING-FOOD WEB | ALONGSHORE WIND | EKMAN TRANSPORT OFFSHORE | DEEPER WATER RISES | NUTRIENT SUPPLY | OXYGEN, LIGHT, HABITAT, FISHING PRESSURE

ALONGSHORE WIND → Ekman transport offshore → DEEPER WATER RISES → nutrient supply → PHYTOPLANKTON → zooplankton → fish → LIMITS → oxygen, light, habitat, fishing pressure

DEFINITION / COMPONENTS
- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

TRIGGER / MECHANISM
- PHYTOPLANKTON → zooplankton → fish
- LIMITS → oxygen, light, habitat, fishing pressure

EFFECT / INTERVENTION
- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PHYTOPLANKTON → zooplankton → fish; LIMITS → oxygen, light, habitat, fishing pressure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WESTERN LIMB → narrow, swift, warm; EASTERN LIMB → broad, cool, upwelling-prone.

03

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ALONGSHORE WIND → Ekman transport offshore → DEEPER WATER RISES → nutrient supply → PHYTOPLANKTON → zooplankton → fish → LIMITS → oxygen, light, habitat, fishing pressure

DEFINITION / COMPONENTS
- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

TRIGGER / MECHANISM
- PHYTOPLANKTON → zooplankton → fish
- LIMITS → oxygen, light, habitat, fishing pressure

EFFECT / INTERVENTION
- ALONGSHORE WIND → Ekman transport offshore
- DEEPER WATER RISES → nutrient supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PHYTOPLANKTON → zooplankton → fish; LIMITS → oxygen, light, habitat, fishing pressure.

04

STAGE 04 TIDE GEOMETRY

TIDE GEOMETRY | FULL MOON | ALIGNED FORCES | SPRING TIDE | LARGER RANGE | QUARTER MOON | PARTIAL OFFSET | NEAP TIDE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
NEW / FULL MOON → aligned forces	QUARTER MOON → partial offset	NEW / FULL MOON → aligned forces
SPRING TIDE → larger range	NEAP TIDE → smaller range	SPRING TIDE → larger range

DEFINITION / COMPONENTS
- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

TRIGGER / MECHANISM
- QUARTER MOON → partial offset
- NEAP TIDE → smaller range

EFFECT / INTERVENTION
- NEW / FULL MOON → aligned forces
- SPRING TIDE → larger range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUARTER MOON → partial offset; NEAP TIDE → smaller range.

05

STAGE 05 LOCAL TIDAL RANGE

LOCAL TIDAL RANGE | ASTRONOMICAL FORCING | INCOMING TIDE | SHELF DEPTH + BAY SHAPE | RESONANCE + FRICTION | TIMING AND HEIGHT | SAME FORCING, DIFFERENT LOCAL RANGE

ASTRONOMICAL FORCING → incoming tide → SHELF DEPTH + BAY SHAPE → amplification → RESONANCE + FRICTION → timing and height → RULE → same forcing, different local range

DEFINITION / COMPONENTS
- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

TRIGGER / MECHANISM
- RESONANCE + FRICTION → timing and height
- RULE → same forcing, different local range

EFFECT / INTERVENTION
- ASTRONOMICAL FORCING → incoming tide
- SHELF DEPTH + BAY SHAPE → amplification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESONANCE + FRICTION → timing and height; RULE → same forcing, different local range.

06

STAGE 06 SALINITY BUDGET

SALINITY BUDGET | EVAPORATION + BRINE | SALINITY RISES | RAIN + RIVER + MELT | SALINITY FALLS | ADVECTION + MIXING | REDISTRIBUTE SALT | TEMP + SALINITY

EVAPORATION + BRINE → salinity rises

RAIN + RIVER + MELT → salinity falls → ADVECTION + MIXING → redistribute salt → TEMP + SALINITY → density and stratification

DEFINITION / COMPONENTS
- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

TRIGGER / MECHANISM
- ADVECTION + MIXING → redistribute salt
- TEMP + SALINITY → density and stratification

EFFECT / INTERVENTION
- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAIN + RIVER + MELT → salinity falls.

06

STAGE 06

SALINITY BUDGET

SALINITY BUDGET | EVAPORATION + BRINE | SALINITY RISES | RAIN + RIVER + MELT | SALINITY FALLS | ADVECTION + MIXING | REDISTRIBUTE SALT | TEMP + SALINITY

EVAPORATION + BRINE → salinity rises

RAIN + RIVER + MELT → salinity falls → ADVECTION + MIXING → redistribute salt → TEMP + SALINITY → density and stratification

DEFINITION / COMPONENTS
- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

TRIGGER / MECHANISM
- ADVECTION + MIXING → redistribute salt
- TEMP + SALINITY → density and stratification

EFFECT / INTERVENTION
- EVAPORATION + BRINE → salinity rises
- RAIN + RIVER + MELT → salinity falls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAIN + RIVER + MELT → salinity falls.

07

STAGE 07

ARABIAN-BAY CONTRAST

ARABIAN-BAY CONTRAST | ARABIAN SEA | STRONGER EVAPORATION | BAY OF BENGAL | RAIN + LARGE RIVER INPUT | GENERAL RESULT | ARABIAN SEA MORE SALINE | SEASON AND LOCAL MIXING MATTER

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
ARABIAN SEA → stronger evaporation	GENERAL RESULT → Arabian Sea more saline	ARABIAN SEA → stronger evaporation
BAY OF BENGAL → rain + large river input	QUALIFIER → season and local mixing matter	BAY OF BENGAL → rain + large river input

DEFINITION / COMPONENTS
- ARABIAN SEA → stronger evaporation
- BAY OF BENGAL → rain + large river input

TRIGGER / MECHANISM
- GENERAL RESULT → Arabian Sea more saline
- QUALIFIER → season and local mixing matter

EFFECT / INTERVENTION
- ARABIAN SEA → stronger evaporation
- BAY OF BENGAL → rain + large river input

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BAY OF BENGAL → rain + large river input.

08

STAGE 08

MONSOON-CURRENT SWITCH

MONSOON-CURRENT SWITCH | SW MONSOON | WINDS DRIVE SUMMER REORGANISATION | SOMALI SECTOR | STRONG CURRENT + UPWELLING | NE MONSOON | NORTHERN FLOWS REVERSE/REORGANISE | MAP RULE

SW MONSOON → winds drive summer reorganisation

SOMALI SECTOR → strong current + upwelling → NE MONSOON → northern flows reverse/reorganise → MAP RULE → label every arrow by season

SYSTEM / DEFINITION
- SW MONSOON → winds drive summer reorganisation
- SOMALI SECTOR → strong current + upwelling

PROCESS / MECHANISM
- NE MONSOON → northern flows reverse/reorganise
- MAP RULE → label every arrow by season

SPATIAL / INDIA PATTERN
- SW MONSOON → winds drive summer reorganisation
- SOMALI SECTOR → strong current + upwelling

HAZARD / LIMIT / RESPONSE
- NE MONSOON → northern flows reverse/reorganise
- MAP RULE → label every arrow by season

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP RULE → label every arrow by season.

09

STAGE 09

IOD COUPLED SEESAW

IOD COUPLED SEESAW | WEST-EAST SST CONTRAST | PRESSURE GRADIENT | THERMOCLINE AND UPWELLING RESPONSE | RAINFALL SHIFTS ZONALLY | INDIA EFFECT | CONDITIONAL ON TIMING + ENSO + MJO

WEST-EAST SST CONTRAST → pressure gradient

WINDS → thermocline and upwelling response → CONVECTION → rainfall shifts zonally → INDIA EFFECT → conditional on timing + ENSO + MJO

SYSTEM / DEFINITION
- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

PROCESS / MECHANISM
- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

SPATIAL / INDIA PATTERN
- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

HAZARD / LIMIT / RESPONSE
- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INDIA EFFECT → conditional on timing + ENSO + MJO.

09

STAGE 09

IOD COUPLED SEESAW

IOD COUPLED SEESAW | WEST-EAST SST CONTRAST | PRESSURE GRADIENT | THERMOCLINE AND UPWELLING RESPONSE | RAINFALL SHIFTS ZONALLY | INDIA EFFECT | CONDITIONAL ON TIMING + ENSO + MJO

WEST-EAST SST CONTRAST → pressure gradient

WINDS → thermocline and upwelling response → CONVECTION → rainfall shifts zonally → INDIA EFFECT → conditional on timing + ENSO + MJO

SYSTEM / DEFINITION
- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

PROCESS / MECHANISM
- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

SPATIAL / INDIA PATTERN
- WEST-EAST SST CONTRAST → pressure gradient
- WINDS → thermocline and upwelling response

HAZARD / LIMIT / RESPONSE
- CONVECTION → rainfall shifts zonally
- INDIA EFFECT → conditional on timing + ENSO + MJO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INDIA EFFECT → conditional on timing + ENSO + MJO.

10

STAGE 10

OBSERVATION-TO-SERVICE CHAIN

OBSERVATION-TO-SERVICE CHAIN | OCEAN + WEATHER OBSERVATIONS | DATA MANAGEMENT AND PRODUCTS | ANALYSIS AND FORECAST ENSEMBLES | MARINE SERVICE WITH UNCERTAINTY

OMNI / RAMA → ocean + weather observations

INCOIS → data management and products → MODELS → analysis and forecast ensembles → USER → marine service with uncertainty

DEFINITION / COMPONENTS
- OMNI / RAMA → ocean + weather observations
- INCOIS → data management and products

TRIGGER / MECHANISM
- MODELS → analysis and forecast ensembles
- USER → marine service with uncertainty

EFFECT / INTERVENTION
- OMNI / RAMA → ocean + weather observations
- INCOIS → data management and products

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OMNI / RAMA → ocean + weather observations.

11

STAGE 11

OCEAN ANSWER SPINE

OCEAN ANSWER SPINE | DEPTH, SCALE AND SEASON | FORCE, DENSITY OR GRAVITY | BASIN, CURRENT, TIDE OR SALINITY | VARIABILITY, ECOLOGY AND EVIDENCE DATE

DEFINE → depth, scale and season → DERIVE → force, density or gravity → MAP → basin, current, tide or salinity → QUALIFY → variability, ecology and evidence date

DEFINITION / COMPONENTS
- DEFINE → depth, scale and season
- DERIVE → force, density or gravity

TRIGGER / MECHANISM
- MAP → basin, current, tide or salinity
- QUALIFY → variability, ecology and evidence date

EFFECT / INTERVENTION
- DEFINE → depth, scale and season
- DERIVE → force, density or gravity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → basin, current, tide or salinity; QUALIFY → variability, ecology and evidence date.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | THE ARABIAN SEA HOSTS ONE OF THE WORLD'S STRONGEST | WHY HERE | MONSOON-DRIVEN PRODUCTIVITY AND ORGANIC-MATTER DECAY COMBINE WITH WEAK VENTILATION. | LOSS OF FIXED NITROGEN VIA DENITRIFICATION/ANAMMOX , RELEASE OF | FISHERY SHIFTS, AND REPLACEMENT OF DIATOMS BY HYPOXIA-TOLERANT NOCTILUCA | MAJID HUSAIN LINKS ABNORMALLY HIGH SEA TEMPERATURES WITH CORAL | MARINE HEATWAVE

OPTIONAL PROCESS DEPTH
- CAUTION: The Arabian Sea hosts one of the world's strongest tropical oxygen-minimum zones at
- CAUTION: Why here: monsoon-driven productivity and organic-matter decay combine with weak ventilation.
- CAUTION: Consequences: loss of fixed nitrogen via denitrification/anammox , release of the greenhouse gas nitrous oxide (N_2O) ,
- fishery shifts, and replacement of diatoms by hypoxia-tolerant Noctiluca (green Noctiluca blooms) — an India-relevant fisheries/climate concern.

DATA / INSTITUTION LIMIT
- Majid Husain links abnormally high sea temperatures with coral bleaching, where heat stress kills the microscopic algae that colour and feed corals.
- CAUTION: Marine heatwave = a prolonged spell of unusually high sea-surface temperature
- over a specific ocean region, relative to its normal seasonal range.
- CAUTION: Mechanism: ocean warming + weak winds/mixing + persistent high pressure can suppress cooling and keep surface waters anomalously warm.

CONTEMPORARY PARALLEL
- CAUTION: India link: Arabian Sea and Bay of Bengal warming can intensify cyclones, stress corals and alter fisheries/monsoon moisture supply.
- North Indian Ocean currents reverse seasonally with the monsoon.
- Summer = SW Monsoon Drift; winter = NE Monsoon Drift.
- South Indian Ocean has a normal subtropical gyre.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels